CCIS Guideline for Teaching Slide Decks

Oct 22, 2026 · @Duhai Alshukaili

Purpose and scope

A teaching slide should help students think, not serve as the lecturer's script. This guideline sets a common minimum standard for slide decks used in CCIS courses across all UTAS branches, covering ten areas from layout to citation.

The recommendations draw on the background study *Pedagogical Slide Design in Higher Education*, which synthesizes Cognitive Load Theory (Sweller), Mayer's Cognitive Theory of Multimedia Learning, the Assertion-Evidence framework (Alley and Garner), and WCAG 2.2 accessibility standards. Where the evidence is strong, the guideline says "must"; where it is closer to good practice or depends on the course, it says "should" or "consider."

Each section gives a short rule, then a **Good** and **Bad** example side by side. Faculty adapting existing decks do not need to rebuild everything at once; Sections 4, 8 and 9 (branding, attribution, citations) are the priority because they carry institutional and legal weight.

1. Clear and simple design

One slide carries one idea. Working memory can only handle a few new elements at once, so every decorative border, animation or extra bullet competes with the concept you are teaching (the coherence principle). Reading dense on-screen text aloud makes things worse, since students then process two overlapping verbal streams (the redundancy effect).

For most technical content, CCIS recommends the **Assertion-Evidence** layout: a headline that is a full sentence stating the point (one to two lines, roughly 12 to 20 words), and a body made of visual evidence such as a diagram, code fragment, plot or worked example. Bullet lists are kept for logistics and recaps.

	Good	Bad
Headline	"Binary search halves the search space at each step, so it runs in O(log n) time"	"Binary Search"
Body	A sorted array with the low, mid and high pointers drawn across three iterations	Eight bullets copied from the textbook definition
Layout	Generous white space; content aligned to a grid	Every corner filled; text boxes at random positions
Delivery	Lecturer explains the diagram verbally	Lecturer reads the bullets word for word
Animation	A build that reveals one step of an algorithm at a time	Spinning or bouncing entrances on every bullet

Rules of thumb:

- If a slide needs more than about 40 words of body text, split it or move the text to the speaker notes.
- Put the detailed explanation in the notes pane and in the handout, not on the projected slide.
- Use built-in layout placeholders (title, content) rather than floating text boxes; screen readers depend on them.

2. Proper use of graphics

A graphic earns its place only if it explains something the words cannot. Diagrams, architecture sketches, plots, state machines and annotated screenshots are evidence; clip art and generic stock photos are noise. Labels belong next to the part they describe (spatial contiguity), and arrows, highlights or bolding should steer the eye to the one element that matters (signaling).

	Good	Bad
Choice of image	TCP three-way handshake drawn as a sequence diagram with labeled SYN, SYN-ACK and ACK	A stock photo of a handshake between two business people
Labels	Labels placed on each layer of the OSI diagram	A numbered diagram with a separate legend at the bottom of the slide
Emphasis	The slow query highlighted in one accent color within the execution plan	Five colors used with no explanation of what each means
Resolution	Vector diagrams (SVG, TikZ, native shapes) or high-resolution images	A blurry, stretched screenshot copied from a PDF
Code	10 to 15 lines in a monospaced font at 20 pt or more, with the key line highlighted	A full page of code at 10 pt
Accessibility	Alt text stating the point: "Bar chart showing that model accuracy plateaus after 50 epochs"	No alt text, or "image1.png"

AI-generated images may be used, but the lecturer is responsible for checking them for technical errors and for stereotyped depictions of people before they reach students. Decorative items (background shapes, the logo) should be marked as decorative so screen readers skip them.

3. Font choice

Use one sans-serif family for the whole deck, plus one monospaced font for code. Projected text has to survive glare, low-resolution projectors and the back row; thin serif strokes tend to break down under those conditions. Fonts that clearly separate I, l and 1 (and O and 0) matter in a computing course where students copy identifiers and code.

Element	Recommended size and style
Slide headline	32 to 40 pt, bold, sentence case

Element	Recommended size and style
Body text, diagram labels, callouts	20 to 24 pt, regular or medium
Captions, axis labels, source lines	14 to 16 pt minimum
Code	18 to 22 pt monospaced (e.g., Consolas, Fira Mono, Source Code Pro)

	Good	Bad
Typeface	Calibri, Arial, Roboto or Atkinson Hyperlegible throughout	Times New Roman body text with a script font for headings
Consistency	Same font and sizes on every slide, set in the master	Each slide formatted by hand, sizes drifting from 18 to 28 pt
Size	Smallest text on screen is 14 pt	9 pt footnotes and 12 pt table cells
Case and emphasis	Sentence case; bold for the one key term	ALL CAPS headings, underlined and italic at once
Alignment	Left-aligned text, line spacing 1.3 to 1.5	Fully justified text with uneven gaps between words

When a slide mixes Arabic and English, choose a family with proper Arabic glyphs (for example Arial or Noto Sans Arabic) so that the two scripts do not fall back to mismatched fonts.

4. University branding

Every deck opens with a title slide that carries the official UTAS logo, taken from the high-quality PNG file shared with all staff. The logo is not downloaded from the web, redrawn or recolored.

The official UTAS PowerPoint template is **not** recommended for teaching. Its colorful margins and wide borders take space away from content and add the kind of visual noise the background study warns against. Instead, faculty build a **course template**, ideally as a joint effort among the lecturers who teach the same course across branches. The key requirement is consistency: one template is used for every slide of every lecture in the course.

A course template should include:

- A title slide layout with the UTAS logo, college name, course code and title, lecture number and topic, lecturer and branch, and semester
- A section divider layout
- A content layout with a headline area and a large, plain content area with narrow margins
- A slim footer: course code, lecture number, slide number
- Fixed fonts and sizes (Section 3) and a small palette: one text color plus one or two accents (Section 5)
- A closing layout for the summary and references

One lecturer (for example the course coordinator) keeps the master copy in a shared folder, and changes are agreed by the course team rather than made separately in each branch.

	Good	Bad
Title slide	Official UTAS logo from the staff PNG file, undistorted, with full course details	No logo, or a logo copied from a web search
Logo quality	PNG used at or below its original size, aspect ratio locked	Logo stretched, enlarged until pixelated, recolored or placed on a clashing background
Template choice	A plain course template with narrow margins, agreed by the course team	The corporate UTAS template with colorful borders used for lectures
Consistency	All lectures of the course, at every branch, use the same template	Lecture 1 in one theme and Lecture 5 in another; each branch with its own look
Content slides	Slim footer with course code, lecture and slide number	Large logo and banner repeated on every slide
Other logos	Partner or vendor logos only where a formal partnership exists (e.g., a certification track)	Company logos used as decoration

If the logo also appears on content or closing slides, keep it small, outside the content area, and marked as decorative in its alt-text settings.

5. Choice of colors

Use a restrained palette: a neutral background, one dark text color, and one or two accent colors defined once in the course template (Section 4). Color is a signal, so it should mark what matters rather than decorate.

Two checks are required:

1. **Contrast.** Text must reach a contrast ratio[1] of at least 4.5:1 against its background (3:1 for large text of 18 pt, or 14 pt bold). Classrooms with bright light or older projectors justify aiming higher, at 7:1. Free checkers (for example the WebAIM contrast checker or the built-in accessibility checker in PowerPoint) take a few seconds to run.
2. **Color is never the only cue.** About 8% of men and 0.5% of women have some form of color vision deficiency, most often red-green. In a class of 30 students split evenly between men and women, this means roughly one affected student on average, and about a 73% chance that at least one student in the room cannot reliably tell red from green. In an all-male section of 30 the chance rises above 90%; in an all-female section it is about 14%. A red-versus-green chart will, in most mixed classes, fail for at least one student. Pair colors with shapes, line styles or labels.

[1] *What the ratio means.* WCAG compares the relative luminance (perceived brightness) of the lighter color, L1, with that of the darker color, L2. Luminance runs from 0 for pure black to 1 for pure white:

```
\text{Contrast ratio} = \frac{L_1 + 0.05}{L_2 + 0.05}
```

The result ranges from 1:1 (text the same color as its background, invisible) to 21:1 (black on white). For reference, black on white is 21:1, mid-gray #767676 on white is about 4.5:1 (just passing), and yellow on white is about 1.1:1 (fails). The 0.05 added to both values accounts for ambient light reflected off the screen. The checkers mentioned above compute this automatically.

	Good	Bad
Background and text	Dark gray or black text on white or off-white	Light gray text on white, or yellow text on light blue

	Good	Bad
Accent	One accent color highlights the changed line in a code diff	Every keyword in a different color
Charts	Two series as a solid line with circles and a dashed line with triangles, each labeled directly	Two series distinguished only by red and green
Backgrounds	Plain background	Text placed over a busy photo or gradient
Dark theme	Used deliberately for code demos in a dimmed room, with contrast checked	Dark slides projected in a bright room, so text glows and blurs

Code syntax highlighting is acceptable, but pick a theme with good contrast on the projector and keep it the same across the course.

6. Software tools: LaTeX, PowerPoint, Google Slides

Any of the three is acceptable. The choice should follow the content and the way the deck will be maintained, not habit. Whatever the tool, every lecture in a course must use the same course template (Section 4) and be shared with students as a PDF on the LMS unless editing is part of the activity.

Tool	Best suited to	Watch out for
LaTeX (Beamer)	Mathematics-heavy courses (algorithms, theory, discrete math), consistent typesetting, code listings, version control with Git	Default themes are cluttered (navigation bars, mini-frames); exported PDFs are often poorly tagged for screen readers; the course template has to be rebuilt as a plain Beamer theme with the same layout and logo
PowerPoint	Most courses; diagram-heavy lectures; strongest built-in accessibility checker, alt-text and reading-order tools	Ad-hoc text boxes instead of placeholders; overuse of transitions; large files when images are not compressed

Tool	Best suited to	Watch out for
Google Slides	Courses co-taught across branches where several lecturers edit one deck; quick sharing	Fewer accessibility tools; fonts may change on export; edit access must be controlled so students receive a fixed version

	Good	Bad
Template	Every lecture starts from the shared course template, in the tool the course team agreed on	Lecturer starts from a blank slide or a random online theme
Accessibility	Runs Review > Check Accessibility in PowerPoint before uploading	Uploads without any check
Beamer	Uses a plain theme, `\section{}` for structure and the `listings` or `minted` package for code	Uses a theme with a navigation bar and sidebar on every frame and pastes code as images
Shared decks	One master copy per course in a shared drive, with version history	Each branch keeps its own edited copy with no record of changes
Distribution	PDF uploaded to the LMS, with an accessible version on request	Editable source shared publicly, or slides sent only through messaging apps

When branches teaching the same course prefer different tools, the course team should agree on one tool for the shared template. Parallel Beamer and PowerPoint versions of one course tend to drift apart within a semester or two.

7. Information that goes on the slides

Certain information is required on fixed slides; everything else is decided by the learning outcome of the lecture.

Required on the title slide: UTAS logo; College of Computing and Information Sciences; course code and title; lecture number and topic; lecturer name (and branch, where relevant); semester and academic year.

Required near the start: the learning outcomes of the lecture, written with measurable verbs and mapped to the course learning outcomes (for example "Trace the execution of Dijkstra's algorithm on a weighted graph," not "Understand shortest paths").

Required at the end: a short summary or recap, a pointer to the next lecture or the related lab, and the reference list (see Section 9).

On content slides: the claim, the evidence for it, and the source. Explanations, extra examples and background go in the speaker notes or the handout.

	Good	Bad
Title slide	"COMP3105 Information Retrieval · Lecture 4: Vector Space Model · Dr. A. Al-Hinai · UTAS-Ibri · Fall 2026"	"Chapter 4" with no course, date or lecturer
Outcomes	Three measurable outcomes on slide 2	No outcomes, or outcomes copied from the whole course spec
Content	Headline claim, one worked example, source line	A transcript of everything the lecturer intends to say
Activities	A slide with a short problem or poll question every 15 to 20 minutes	60 content slides with no pause for students
Personal data	Anonymized sample data and screenshots	Student names, IDs, marks or real login credentials visible in screenshots

The example title line above uses invented details for illustration only.

8. Attributing others' contributions

If a deck builds on slides made by someone else, that contribution is acknowledged on the slides themselves. CCIS penalizes students for unacknowledged borrowing, and the same standard applies to faculty. Sharing slides between colleagues and branches is encouraged; leaving the original author out is not.

What to do:

- Add an **acknowledgment line on the title slide** or a short credits slide right after it: whose slides were used, which course or edition, and what you changed.
- Keep the original author's credit on any individual slide reused unchanged.

- Check the terms first. Publisher instructor slides are usually licensed only to adopters of that textbook; slides from other universities' open courses may carry a Creative Commons license with conditions (for example NonCommercial or ShareAlike) that must be respected.
- If generative AI helped draft the deck, say so in one line (for example "Draft headlines and practice questions prepared with an AI assistant; content reviewed and verified by the lecturer"). The lecturer remains responsible for every claim on the slides.

	Good	Bad
Colleague's slides	"Adapted from slides by Dr. X (UTAS-Nizwa, Fall 2025); examples and lab exercises revised by Dr. Y"	Colleague's name removed and replaced with your own
Textbook slides	"Based on the instructor slides accompanying [textbook, edition], used under the publisher's adopter terms"	Publisher logo cropped out and slides presented as original
Open course slides	"Adapted from [course name, university], licensed CC BY-NC-SA 4.0; this adaptation is shared under the same license"	CC-licensed slides re-shared with no license line
AI assistance	Short disclosure line on the credits slide	No disclosure, or AI-generated content passed off as checked when it was not

The colleague, branch and course names in the examples are placeholders.

9. Referencing and citations

Every figure, table, dataset, quotation or code sample taken from elsewhere is cited **on the slide where it appears**, in a source line of at least 14 pt beneath it, and listed in full on a References slide at the end. A single reference list hidden in 8 pt type on the last slide does not tell students where any particular figure came from.

APA 7 is the recommended citation style for CCIS slide decks. Faculty who prefer IEEE may use it, provided the whole course uses that one style consistently; mixing styles within a deck or across lectures of the same course is not acceptable. For openly licensed images, the source line follows the TASL pattern: **T**itle, **A**uthor, **S**ource (link), **L**icense.

	Good	Bad
Figure from a book (APA 7)	Under the figure: "Note. Adapted from *Introduction to Information Retrieval* (p. 121), by C. D. Manning, P. Raghavan, & H. Schütze, 2008, Cambridge University Press." References slide: Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to information retrieval*. Cambridge University Press.	Figure with no source, or "Source: Google"
Consistency	APA 7 throughout the course (or IEEE throughout, if the course team chose it)	APA in-text citations on some slides and IEEE numbers [3] on others
Open image	"'Network topology diagram' by J. Smith, Wikimedia Commons, CC BY-SA 4.0" (with links)	Image copied from a search result with no license check
Web source	Author, date, title and URL (or DOI) on the References slide, as APA 7 requires	A shortened link or "from the internet"
Code	"Adapted from the scikit-learn documentation example, BSD license"	Stack Overflow answer pasted without a link
AI output	Every reference produced or suggested by an AI tool is checked against the original source before use	AI-suggested references used unchecked; some do not exist

The page number in the book example is illustrative; use the actual page or figure number of the source.

A note on copyright: the background study's legal analysis rests on US law (classroom exemption, TEACH Act, fair use). Those doctrines do not transfer automatically to Oman. Showing a figure live in class and uploading it to the LMS are different acts, so faculty

should prefer openly licensed or self-made figures, use only the portion of a work that is needed, and follow Omani copyright law and UTAS policy for anything published beyond the classroom.

10. Sectioning and transitioning between topics

A lecture deck should have a visible structure of three to five sections, each tied to one of the lecture's outcomes. Students lose the thread when topics change without warning; a short section divider gives them a moment to close one idea and open the next.

Recommended skeleton for a 50 to 90 minute lecture:

1. Title slide (with logo and acknowledgments, Sections 4 and 8)
2. Learning outcomes, and a one-line link to the previous lecture
3. Agenda listing the sections
4. Section divider, then content slides, ending with a check-for-understanding question; repeat for each section
5. Summary that restates the main assertions
6. Next steps: lab, reading, or assessment due
7. References

	Good	Bad
Dividers	A plain divider slide "Part 2 of 3: Indexing" with the agenda shown and the current section highlighted	40 slides in a row with no visible break between topics
Bridging	The last slide of a section asks "Linear scan is O(n). Can we do better?" and the next section answers it	The next topic starts abruptly with a new definition
Slide transitions	None, or a simple fade used the same way throughout	Cube, vortex or random transitions between slides
Software structure	PowerPoint Sections, Beamer `\section{}`, or Google Slides section headers, so the outline is navigable	Structure exists only in the lecturer's head

	Good	Bad
Recap	Summary slide repeats the three headline assertions from the lecture	"Any questions?" as the only closing slide

Checklist before uploading a deck

- [ ] Same course template on every slide; UTAS logo (official PNG) on the title slide
- [ ] Title slide has course code, lecture topic, lecturer, branch, semester
- [ ] Learning outcomes stated with measurable verbs
- [ ] One idea per slide; headlines state the point
- [ ] No text below 14 pt; one sans-serif family plus one code font
- [ ] Contrast checked (4.5:1 minimum); no meaning carried by color alone
- [ ] Every informative image has alt text; decorative items marked as decorative
- [ ] Clear sections with dividers, a summary and next steps
- [ ] Others' slides acknowledged; license terms respected; AI assistance disclosed
- [ ] Every borrowed figure, table and code sample cited on its slide in one consistent style (APA 7 recommended); References slide complete
- [ ] No student personal data in screenshots or examples
- [ ] Accessibility checker run; PDF exported for the LMS

Sources

All recommendations above are drawn from the background study *Pedagogical Slide Design in Higher Education: Cognitive Architecture, Visual Aesthetics, Academic Integrity, and Generative AI Integration*, in particular its treatment of:

- Cognitive Load Theory (Sweller) and Mayer's Cognitive Theory of Multimedia Learning (coherence, signaling, spatial contiguity, redundancy and modality principles)
- The Assertion-Evidence framework (Alley and Garner) and Garr Reynolds' guidance on signal-to-noise and white space
- WCAG 2.2 contrast, color-independence and alt-text requirements
- Creative Commons licensing and the TASL attribution pattern
- The Russell Group (2023) and UNESCO (2023) guidance on generative AI in education